

Fundamentals of Renewabale Energy

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

6104B

Credits

4

Periods

5/week

Course Details

Title: Fundamentals of Renewabale Energy

Department: Common

Revision: REV_Fundamentals of Renewabale Energy

Pattern: Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 6104B — Fundamentals of Renewabale Energy**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ഐഐഐഐഐ ഐഐഐഐ ഐഐഐഐ English technical terms- ഐഐഐഐഐഐഐ. ുഐ ഐഐഐഐ ഐഐഐഐ ഐഐഐഐഐഐഐ, formula, diagram, units, definitions ഐഐഐഐ ഐഐഐഐഐഐഐ ഐഐഐഐഐഐഐ.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Fundamentals of Renewable Energy.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Fundamentals of Renewable Energy.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഏറ്റവും അടിസ്ഥാനപരമായ പദപരിചയം നൽകുന്നതിനും വിദ്യാർത്ഥികൾക്ക് ആവശ്യമായ അടിസ്ഥാനപരമായ അറിവുകൾ നൽകുന്നതിനും.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: പ്രായോഗികമായി ഉപയോഗിക്കാവുന്ന കണക്കുകൂട്ടൽ രീതികൾ പഠിപ്പിക്കുന്നതിനും വിദ്യാർത്ഥികൾക്ക് ആവശ്യമായ പ്രായോഗിക പ്രക്രിയകൾ പഠിപ്പിക്കുന്നതിനും.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഐക്യകേരളം ഐക്യമേഖലയായി മാറട്ടെന്ന് ഉറപ്പാക്കുന്നു.
 ഐക്യകേരളം ഐക്യമേഖലയായി മാറട്ടെന്ന് ഉറപ്പാക്കുന്നു.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: മലയാളത്തിൽ ഈ സോഫ്റ്റ്‌വെയർ പ്രവർത്തിക്കുന്നില്ല.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Fundamentals of Renewable Energy.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.